HAOYAN ZHENG

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EDUCATION

Bachelor of Applied Science and Engineering (B.A.Sc) in Engineering Science Sep. 2022 - Jun. 2026
University of Toronto, Toronto, ON

- **Major** in **Robotics Engineering, Minor** in **Artificial Intelligence**
- **Relevant Courses**: Computer Algorithms and Data Structures (Python, C/C++), Digital System and Computer Organizations (System Verilog, Assembly), Electromagnetism, Microprocessors and Embedded Microcontrollers, Unsupervised Machine Learning.
- **Software:** ModelSim, Quartus, Git, Visual Code Studio, AutoCAD, Fusion, LTspice, Matlab, Covidence, Microsoft Office
- **Languages:** Python, Java, C/C++, Matlab, System Verilog, Assembly, ROS

EXPERIENCE

Research Analyst Intern May. 2024 - Present
Michael Garron Hospital, East York, ON

- Contributed to Agency for Healthcare Research and Quality (AHRQ) funded Diagnostic Center of Excellence on **diagnostic and human-factor research**.
- Conducted scoping reviews by searching peer-reviewed and grey literature on **PubMed, Scopus, MEDLINE, and Embase**.
- Screened, reviewed, and extracted more than **10,000 peer-reviewed publications** using Covidence.
- Assisted in developing **Review Ethics Board (REB) protocol and supplemental documents** such as recruitment scripts, consent forms, and data collection instruments for patient interviews.
- Designed activities for **World Health Organization (WHO) Patient Safety Day** to engage patients, family, and healthcare professionals to increase awareness of patient safety in diagnosis.

PROJECTS

Patient Survey Analysis Using Unsupervised Machine Learning *Python, nltk, Gensim, hyperopt, guidedlda*

- Developed an unsupervised Natural Language Processing (NLP) machine learning algorithm using **Latent Dirichlet Allocation (LDA) Topic Modelling** to **automatically** identify themes from patient free-text survey responses in less than 0.05 s.
- Optimized 10 **hyperparameters** of the LDA model using **Adaptive Tree of Parzen Estimators (TPE)**, a Bayesian optimization method to maximize the **coherence and perplexity** scores of the model.
- Developed comprehensive text pre-processing steps including **stop word removal, spelling check, text cleaning, POS tagging, lemmatization, and tokenization**.
- Trained, optimized the final LDA model using a training set of **1,500 patient free-text survey responses** and evaluated the Topic Model using evaluation metrics.

Autonomous Delivery Robot *ROS, Python*

- Designed and implemented a **Bayesian localization-based control system** for a TurtleBot3 robot to autonomously deliver mail to three randomized offices in a quadrilateral hallway.
- **Leveraged ROS, computer vision, and line-following algorithms** to achieve **accurate localization** in a multi-colored environment with only four unique colors.
- Designed Python robot simulation code for **measuring, planning, and acting** to test the proof-of-concept.
- Successfully completed mail delivery tasks running at $0.1m/s$, demonstrating robust navigation on a closed-loop route.

ACHIEVEMENTS

UTMIST Hackathon - Winner *Mar. 2023* University of Toronto
Honored as the winning team in the University of Toronto Machine Intelligence Student Team (UTMIST) Hackathon.

2022 Physics Olympics - 1st Place *Mar. 2022* University of British Columbia
Led a team of **8 International Baccalaureate (IB) students**, outperformed **over 40** competing schools in British Columbia.

PUBLICATIONS

Reimaging Health Equity, Equality, and Justice: A Fresh Vision. Longwoods Insights.
McCallum, E; Scala, N; Mason, T; ***Zheng, H****; Priore, A; Coppinger, T; Sochaniwskyj, M; Pettit A; Hill, MA; Shearkhani, S. (2024).*
longwoods.com/content/27405

How Gender Affects the Leading Causes of Death: What You Need to Know
Zheng, H *(2024)* Patient-Partnered Diagnostic Center Of Excellence
Under Review

POSTERS

Using Natural Language Processing (NLP) to Uncover Patient Insights from Free-text Survey Responses
Diagnostic Excellence 2024 Meeting (DEX24) at the University of Minnesota and UCSF
Zheng, H*; Ni, K; Alfred, M; Smith, K. (2024)*
z.umn.edu/DEX2024

Using Natural Language Processing (NLP) to Uncover Patient Insights from Free-text Survey Responses
Inter-University Workshop (IUW2024) at the State University of New York
Zheng, H*; Ni, K; Alfred, M; Smith, K. (2024)*
ubhfes.wixsite.com/iuw2024